

Literature Review: Indian Market Implementation (IIMA Study)

While the Fama–French and Carhart multifactor models were established for the United States markets, applying them to emerging countries like India needs cautious adjustments because of structural variations such as high market concentration, dominance by large companies, and fiscal year alignment. The Indian Institute of Management Ahmedabad (IIMA) conducted an extensive study (1994–2013) to modify and validate the Fama–French and Carhart models in the Indian context.

First, the factor construction process was adapted to accommodate local market conditions. Initially, the breakpoint size was fixed at the 90th percentile of market cap rather than the U.S. median (50th percentile) due to the high density of large-cap companies in Indian markets. Subsequently, portfolio formation dates were changed to September (rather than June in the United States) to coincide with India's March fiscal year-end accounting cycle. Third, to control for survivorship bias and the common disappearance of troubled companies from Indian markets, a -100% return was used for companies that delisted or went out of business, providing an accurate representation of investor experience.

Applying these modifications, the Indian research developed factor portfolios and examined the explanatory ability of the market, size (SMB), value (HML), and momentum (WML) factors. The model employed the same functional form as that of the Carhart (1997) specification:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i (R_{M,t} - R_{f,t}) + s_i \cdot \text{SMB}(t) + h_i \cdot \text{HML}(t) + p_i \cdot \text{WML}(t) + e_{i,t}$$

Key results from 1994–2013 reveal significant differences from developed economies. The market risk premium (MRP) in India averaged 11.5% per year, indicating higher equity risk than in the U.S. Interestingly enough, the size premium (SMB) proved to be statistically insignificant, averaging around 0% per annum, indicating that size does not have any explanatory power in the case of Indian stock returns. On the other hand, the value effect (HML) was extremely powerful, averaging at 15.3% per annum, reaffirming that high book-to-market (value) companies always outperformed growth companies. Even more impressive was the momentum effect (WML), averaging 21.9% per annum, rendering momentum the prevailing force in India and much more robust than in the U.S. context.

In general, the IIMA study illustrates that although the general structure of multifactor models holds true, local calibration is essential for emerging economies. For India, the value and momentum factors prevail over return variation, whereas the conventional size effect vanishes. The findings are especially significant for portfolio construction, active fund management, and risk modeling in Indian equity markets since they underpin the necessity for investors to depend on value and momentum strategies more than size-based exposures.